

VESPA portal: simple data access to EPN-TAP data

Locate individual data products meeting certain conditions (observational or instrumental)

Typical request for surfaces:

Mars, a given region (Tharsis volcanoes area) — images, spectra or spectral-cubes

Illumination conditions ($i \leq 20^\circ$, phase $\leq 50^\circ$)

Local time or season range (L_s)

Minimum resolution

Spectral range / filter

an instrument

etc

The portal displays a simple parameter form to build the query

Returns results from each service
=> need to be described at similar level

The VESPA portal is a simplified, single-entry point to all validated data services:

<https://vespa.obspm.fr>

All results in a single table

Examine service results

Save on disk

Send to VO apps

The screenshot shows the VESPA portal interface. On the left, there is a 'Refine your search' section with various input fields: Target Name, Target Class, Dataproduct Type, Instrument Host Name, Instrument Name, Processing level, Time, Location, Spectral, Illumination, Data Reference, and Optional. Below these fields are 'Reset' and 'Submit' buttons. On the right, there is a 'Data Services' table listing various data services and their result counts. Each row in the table has three icons: a yellow circle, a blue circle, and a red circle. A purple arrow points from the 'Examine service results' text to the table. Another purple arrow points from the 'Save on disk' text to the yellow circle icon. A third purple arrow points from the 'Send to VO apps' text to the blue circle icon. Below the table, there is a 'Generated WHERE clause of ADQL statement:' section with a complex SQL query. At the bottom of the table, there is a row labeled 'EPN-TAP compilation results'.

Data Services	Results
abs_cs - Data for numerical modeling of planetary atmospheres	13 results
APIS - Auroral Planetary Imaging and Spectroscopy	74175 results
BaseCom - The Nançay Cometary Database	6886 results
bass2000 - Bass2000 solar survey archive	357033 results
BDIP - IAU database of historical planetary images	16906 results
cassini_jupiter - Cassini RPWS/HFR Calibrated Jupiter Flyby Dataset	7 results
CLIMSO - CLIMSO coronagraphs at Pic du Midi de Bigorre	1021345 results
cpstasm - CLUSTER STAFF-SA Spectral Matrix Data	11688 results
DynAstVO - Asteroid orbital database and ephemerides	29733 results
eit_syn - Synchronous synoptic maps of the solar corona from EIT/SoHO	18482 results
ExoPlanet - Extrasolar Planets Encyclopaedia	5177 results

Generated WHERE clause of ADQL statement:

```
SELECT * FROM ... WHERE ((c1min <= c1max AND c1min <= 265.0 AND c1max >= 235.0) OR (c1min > c1max AND (c1min <= 265.0 OR c1max >= 235.0))) AND c2max >= -15.0 AND c2min <= 15.0 AND (1 = ivo_hashlist_has(lower("target_name"),lower('Mars')) OR 1 = ivo_hashlist_has(lower("target_name"),lower('4')) OR 1 = ivo_hashlist_has(lower("target_name"),lower('499')) AND "incidence_min" <= 20.0 AND "spatial_frame_type" = 'body'
```

Resulting TAP query

EPN-TAP compilation results